Zachariah E. Button

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PROFESSIONAL SUMMARY

Applied ML researcher specializing in computer vision, image compression, and sensor fusion. Ten years' experience in technical R&D and data science, with a strong background in deep learning, signal processing, and traditional ML methods. Proficient in Python and PyTorch.

EDUCATION

University of Missouri - Kansas City **Jan 2022 - Present**

- **Degree**: Ph.D. in Computer Science
- **Research Areas**: Efficient neural image compression, multi-sensor fusion
- **Expected:** May 2027

Kansas State University **May 2015**

- **Degree**: M.S. in Statistics
- **Thesis**: Bayesian approach to apply and interpret the two-parameter Item Response Model in the context of replicated preference testing

Kansas State University **May 2013**

- **Degree**: Dual B.S. in Mathematics and Statistics

COURSEWORK

Computer Science

- Computer Vision
- Deep Learning
- Multimedia Communication
- Design & Analysis of Algorithms
- Knowledge Discovery & Management
- C Programming for Engineering (C++)
- Fundamentals of Programming (Java)

Mathematics

- Optimization for Machine Learning
- Analytical Geometry & Calculus I - III
- Applied Matrix Theory (Linear Algebra)
- Advanced Calculus (Real Analysis)
- Foundations of Analysis (Real Analysis)
- Differential Equations
- Algebraic Systems (Number Theory)
- Topology

Statistics

- Statistical Theory I - II
- Statistical Computing (R, SAS)
- Experimental Design for Product Development & Improvement
- Statistical Consulting
- Numerical Methods in Statistics
- Multivariate Statistical Methods
- Theory of Life Data Analysis (Survival Analysis)
- Linear Models
- Applied Linear Statistical Models
- Categorical Data Analysis
- Biostatistics
- Analysis of Variance
- Regression & Correlation Analysis
- Probability & Statistics I - II
- Statistical Methods for Natural Sciences
- Business & Economics Statistics I - II

TECHNICAL SUMMARY

Tools

- **Python** (PyTorch, OpenCV, PySpark, Pandas, scikit-learn, Keras, XGBoost, statsmodels)
- **Deep Learning** (CNN, AE/VAE, RNN, LSTM, R-CNN, Transformers)
- **Databases** (PostgreSQL, Snowflake, Redshift, S3, SQL Server, SQLite, Parquet, HDF5)
- **Compute** (SageMaker, EMR, ECS, Airflow, Slurm)
- **Other** (Cursor, VS Code, Git, Jupyter, Docker, HuggingFace, TensorBoard, W&B, R, Matlab)

Skills

- **Vision** (Classification, Detection, Tracking, Translation, Super-Resolution)
- **Deep Learning** (Optimization, Loss Function Design, Hyperparameter Tuning)
- **Traditional ML** (Regression, Forecasting, Classification, Clustering, Anomaly Detection, Survival Analysis, Dimensionality Reduction)
- **Computation** (Signal Processing, Compression, Entropy Modeling)

EXPERIENCE

Senior Machine Learning Engineer — Ad Astra **May 2026 - Present**

- Own platform-wide ML engineering across 6+ Python repositories, including data pipelines, model development, testing, and production deployment
- Establish best practices and infrastructure for ML development workflows, experimentation, automated testing, code review, production readiness, and cross-team collaboration
- Serve as the company-wide ML thought leader, shaping product strategy and driving the adoption of emerging ML capabilities

Research Engineer — Jumio **May 2025 - Feb 2026**

- Conducted applied research and development for upcoming liveness detection features through an exploration of computer vision and signal processing techniques

Principal Data Scientist — HP, Inc. **Feb 2022 - Mar 2025**

- Led the production lifecycle of machine learning features for an enterprise fleet analytics platform serving **3M** customers, driving a sustained **30%** YoY revenue growth
- Served as the technical leader of a start-up R&D team by setting technical direction, managing cross-functional teams, mentoring junior data scientists, and cultivating a standard of collaboration
- Designed and implemented a scalable risk remediation system based on anomalous device behavior
    - Anomaly Detection stage identifies 3 types of unusual behavior across **20+** device performance metrics and **40M+** consumer devices (PySpark)
    - Risk Modeling stage maps recent anomaly patterns to risk of 4 failure events, using Cox Proportional Hazards models (scikit-survival)
    - Risk Attribution stage combines feature importance metrics with other telemetry data to identify the key factors in increased risk at the device-level
    - Remediation stage maps the key risk factors to suggested actions for reducing failure risk
- Engineered a suite of interpretable models to jointly predict device performance based on hardware and software specifications
    - Foundational models support multiple products, including simulation (effects of configuration changes on device performance) and optimization (recommendations to maximize predicted performance)
    - Two software optimization use cases deployed within 12 months for **80,000** HP devices
    - Tech stack includes SageMaker, EMR, S3, Airflow, GitHub, PySpark, Pandas, and Statsmodels

- Spearheaded the development of an “accelerators” repository, providing utility functions and data processing capabilities to improve the efficiency and accuracy of HP’s AI Systems team

Data Scientist — C2FO **Sep 2019 - Feb 2022**

- Evaluated multiple optimization methods for the platform’s core algorithm, demonstrating potential to increase earned revenue by **$350k/yr** and reduce compute costs by **99%**
- Redesigned customer targeting strategy using a first principles approach and gained support from business leadership
  - Achieved **26%** lift in AUC and **70%** improvement in reliability compared to legacy system
  - Reduced running costs by **75%** and codebase by **80%**
- Developed revenue forecasting system to explain the primary drivers of C2FO market volume, achieving **3x** improvement in monthly budget accuracy

Machine Learning Engineer — Imperial PFS **Nov 2016 - Aug 2019**

- Architected and built vision-based document processing system to enable **$1M+** annual profit increase
  - Methodology included clustering, classification, object detection, OCR, NLP, and expert systems
- Executed controlled A/B testing to evaluate ML-driven customer targeting strategies, resulting in savings of over **$900k** annually
- Automated a large variety of analytics tasks using Python, SQL, and Excel VBA, eliminating **6 hrs/week** of my own work and saving each member of the sales team over **8 hrs/week** in prospecting

Business Intelligence Developer — Epic Systems Corp. **Aug 2015 - Oct 2016**

- Investigated effects of the Affordable Care Act on more than 50 utilization measures using ARIMA forecasting models and intervention analysis
- Contributed to award-winning executive-level utilization dashboard by simulating payment-level claims dataset - demonstrated at HIMSS 2016, Epic UGM 2016, and multiple customer sites

Statistical Consultant — KSU Dept. of Statistics **Aug 2014 - May 2015**

- Advised university researchers on a wide variety of statistical topics, including experimental design, repeated measures analysis, logistic regression analysis, and related interpretations

RESEARCH

University of Missouri - Kansas City **May 2025 - Present**

- Researcher - Multimedia Computing and Communications Lab
  - Z. Button, P. Maharjan, A.M. Badshah, and Z. Li, “SAR-PACT: Tokenization Energy-Aware Complexity Allocation for Practical Learned SAR Image Compression,” *60th Annual Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, 2026.
  - Z. Button, P. Maharjan, and Z. Li, “Accelerating Learned SAR Image Compression via Selective Channel-Group Encoding Bypass,” *IEEE International Conference on Image Processing (ICIP),* Tampere, Finland, 2026.
  - S. Maharjan, R. S. Puttagunta, Z. Button, and Z. Li, "Cross-Modal Thermal Image Compression via RGB Side Information," *IEEE 27th International Workshop on Multimedia Signal Processing (MMSP)*, Beijing, China, 2025.

University of Pittsburgh - Summer Institute for Training in Biostatistics **2013**

- Statistical analysis of adverse cardiac events in patients with Type 2 diabetes or coronary artery disease

Mount Holyoke College - Research Experience for Undergraduates **2012**

- Cluster analysis for tree health in collaboration with the Harvard Forest

TEACHING

Kansas State University — M.S. Data Analytics Program

- Workshop Series Presenter
  - Machine Learning - Classification (GitHub) Mar 2022
  - Data Wrangling (Data.World) Oct 2021
  - Careers in Data Science (Slides) Sep 2020, Sep 2021
  - Machine Learning - Forecasting (GitHub) Oct 2020

Kansas State University — Machine Learning and Autonomous Systems Program

- Advisory Board Member April 2020 - Present
- Instructor Feb 2022 - Apr 2022

Kansas State University — Dept. of Statistics

- Teaching Assistant - Bus. & Econ. Statistics Aug 2013 - May 2015